ATHUL S

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SUMMARY

Electronics and Communication Engineering graduate from Amrita Vishwa Vidyapeetham with a CGPA of 7.08 and a strong STEM foundation. Skilled in Python, data analysis, statistics, and introductory machine learning through the Google Advanced Data Analytics Professional Certificate. Academic and practical experience spans image processing with OpenCV, embedded systems, microcontroller programming, and digital logic design. Seeking to apply analytical problem-solving and rigorous STEM domain knowledge to AI training, model evaluation, and technical content assessment.

WORK EXPERIENCE

Embedded Systems and IoT Training Apr 2025 - Apr 2025
NeST Technologies Kochi, Kerala

- Completed five-day practical training in Arduino and ESP8266-based IoT development.
- Worked on mini-projects involving temperature and motion sensors with Wi-Fi-based data transmission.
- Implemented histogram equalisation using Python and OpenCV for image contrast enhancement.

Industrial Networking and SCADA Exposure Jan 2025 - Jan 2025
KSEB Trivandrum, Kerala

- Gained exposure to SCADA-based monitoring of electrical grid substations.
- Studied industrial communication concepts involving Modbus, TCP/IP and IEC 60870.
- Observed how communication networks support monitoring and telemetry in power-system operations.

EDUCATION

B.Tech - Electronics and Communication Engineering 2022 - 2026
Amrita Vishwa Vidyapeetham, Amritapuri Campus Karunagappally, Kollam
CGPA: 7.08

Higher Secondary (Class XII) 2020 - 2022
Jyothis Central School Kazhakuttam

Secondary School (Class X) 2018 - 2020
Blue Mount Public School Thonnakkal

SKILLS

Programming
Python, Embedded C, Assembly Language

Embedded Systems
ESP32, ESP8266, ATmega328, Arduino IDE, Keil uVision, sensor interfacing

Digital Design
Verilog HDL, Basys 3 FPGA, finite-state machines

Data Analysis & Machine Learning
Python, Statistics, Data Exploration, Predictive Modeling

Electronics & Simulation
Ansys HFSS, MATLAB, antenna parameter analysis

Vision & Industrial Systems
OpenCV, Histogram Equalisation, TCP/IP, Modbus, SCADA

PROJECTS

Chaakra IoT-Based GPS Bus Tracking System Oct 2025 - May 2026

- Developed an academic IoT project using an ESP32 microcontroller and GPS receiver modules for vehicle location tracking.
- Worked with UART communication to receive GPS data and parse NMEA sentences to obtain location coordinates.
- Developed Embedded C firmware for wireless transmission of location data.
- Worked on continuous GPS data acquisition and location updates as part of the tracking system.

Microcontroller Temperature Data Logger Dec 2024 - Dec 2024

- Developed a temperature data logger using an LM35 temperature sensor and ATmega328 microcontroller.
- Wrote Embedded C code for periodic temperature sampling.
- Interfaced a 16x2 LCD to display temperature readings and system status.

Compact Automotive Antenna Design and Simulation Nov 2024 - Jan 2025

- Designed and simulated a compact antenna for 5.8 GHz vehicle-to-vehicle communication using Ansys HFSS.
- Analysed antenna performance using return loss, gain and radiation-pattern simulations.
- Studied the simulated antenna characteristics in relation to the intended V2V application.

FPGA Candy Vending Machine (Basys 3) Feb 2025 - Mar 2025

- Designed a vending-machine control system using Verilog HDL and a Basys 3 FPGA development board.
- Developed modules for finite-state-machine control, button debouncing, BCD conversion and display driving.
- Simulated the control logic to test coin-input handling and item-selection behaviour.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Google Advanced Data Analytics Professional Certificate 2026
Google Career Certificates, Coursera

- Completed an 8-course professional certification covering Python, machine learning, data analysis, statistics, predictive modeling, and data visualization.

NEBOSH International General Certificate (IGC) Result Awaited
Examination completed; result pending.

Machine Learning Training 2024
Amrita Vishwa Vidyapeetham

- Covered Introduction to Machine Learning, Python, data exploration, pre-processing, linear regression, and dimensionality reduction.

INVOLVEMENT

Old Age Home Program Nov 2024 - Nov 2024
Overall Coordinator

- Coordinated social engagement activities for 20+ elderly residents to promote interaction, emotional well-being, and community participation.
- Organized and managed event activities in collaboration with the Student Social Responsibility (SSR) initiative.
- Assisted in coordinating student volunteers and activity planning for community engagement programs involving 20+ participants.